

Practical 9

Questions

25 March 2026

1. Identify the pair of sequences which are close to each other using Hamming and Euclidean distance methods.
(i) AMENLNMDLLYMAAAVMMGLAAIGAAIGIGILGGKFLEGAARQPDLIPLLRTQFFIVMGLVDAIPMI
AVGLGLYVMFAVA
(ii) AADVSAAVGATGQSGMTYRLGLSWDWDKSWWQTSTGRLTGYWDAGYTYWEGGDEGAGKHSLS
FAPVFVYEFAGDSIKPFIEAGIGVAAFSGTRVGDQNLGSSLNFEDRIGAGLKFANGQSVGVRAIHYSNAG
LKQPNDGIESYSLFYKIPI
(iii) MALLPAAPGAPARATPTRWPVGCNRPWTKWSYDEALDGIKAAGYAWTGLLTASKPSLHHATAT
PEYLAALKQKSRHAA
2. Get the non-redundant sequences of beta barrel membrane proteins
(<https://pdbtm.unitmp.org/usage>) with sequence identities of less than 40%, 50%, 75% and 90% using CD-HIT
3. Get the non-redundant sequences of the same type of proteins with sequence identities of less than 20%, 30%, 40% and 50% using PISCES (<https://dunbrack.fccc.edu/piscs/>)
4. Compare the results obtained with the cut-offs 40% and 50%.
5. Extract the non-redundant beta barrel proteins from PDBTM (<https://pdbtm.unitmp.org/usage>) and compare with results from CD-HIT and PISCES using sequence identities of less than 40%.

Deadline: 31st March 2026